

# Bracket Sequences I

**Time limit:** 1.00 s

**Memory limit:** 512 MB

Your task is to calculate the number of valid bracket sequences of length  $n$ . For example, when  $n = 6$ , there are 5 sequences:

- $()()()$
- $()(())$
- $((()))$
- $((()()))$
- $((())())$

## Input

The only input line has an integer  $n$ .

## Output

Print the number of sequences modulo  $10^9 + 7$ .

## Constraints

- $1 \leq n \leq 10^6$

## Example

| Input | Output |
|-------|--------|
| 6     | 5      |